1.

Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

ANS:-

1.Assigning a Static IPv6 Address:

System Used: Windows

Assign a Static IPv6 Address Open Command Prompt as Administrator and run:

```
netsh interface ipv6 set address name="Ethernet" address=2001:db8:1::10
```

2. Verify the Configuration

Run: Ipconfig

The output should show: IPv6 Address: 2001:db8:1::10

Result: The static IPv6 address 2001:db8:1::10 has been successfully assigned and verified.

2.

Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.

ANS:-

Unique Local IPv6 Configuration:

System Used: Windows

Configure a Unique Local IPv6 Address Open Command Prompt as Administrator and run:

```
netsh interface ipv6 set address name="Ethernet" address=fd00:1::10
```

Here, fd00:1::10 is a Unique Local IPv6 address because it starts with fd.

Test Connectivity

Assume the other LAN device has the address fd00:1::20. Run:

```
ping -6 fd00:1::20
```

Result: Successful replies confirm that both devices can communicate using their Unique Local IPv6 addresses.

3.

Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint: Document the steps and the observed address behavior in a text file.

ANS:-

Temporary IPv6 Address:

System Used: Windows

Check the Current IPv6 Address Open Command Prompt and run: Ipconfig

Note down the current IPv6 address of the network adapter.

Disconnect and Reconnect the Network

Disconnect the Wi-Fi/Ethernet connection and reconnect it. Then run:

Ipconfig

again and note the IPv6 address.

Restart the System

Restart the computer and run:

Ipconfig

again to check the IPv6 address.

Observation: After reconnecting the network or restarting the system, the temporary IPv6 address may change. This shows that a temporary IPv6 address is not permanently assigned to the device.

Result: The IPv6 address was checked before and after reconnection/reboot, and its changing behavior was documented.

4.

Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

ANS:-

Zomato-Style Restaurant Search Python Script

Python Script:

```
restaurants = {
    "Dominos Pizza": "2001:db8:1::10",
    "Burger King": "2001:db8:1::20",
```

```
    "Subway": "2001:db8:1::30",
    "Biryani Blues": "2001:db8:1::40"
}

search = input("Enter restaurant name: ").lower()

for name, ipv6 in restaurants.items():
    if search in name.lower():
        print("Restaurant:", name)
        print("IPv6 Address:", ipv6)
```

Example Output:

```
Enter restaurant name: burger
Restaurant: Burger King
IPv6 Address: 2001:db8:1::20
```

5.

Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

ANS:-

Pv6 Subnetting and Manual Configuration

AI-Generated IPv6 Subnetting Scenario:

A small food delivery startup uses the IPv6 network **2001:db8:100::/48**. It is divided into three subnets for different departments:

Department	Subnet
Delivery Partners	2001:db8:100:1::/64
Restaurants	2001:db8:100:2::/64
Admins	2001:db8:100:3::/64

Configuration Steps (Windows):

To manually assign an address from the **Delivery Partners** subnet, open Command Prompt as Administrator and run:

```
netsh interface ipv6 set address name="Ethernet" address=2001:db8:100:1::10
```

Verification:

```
Ipconfig
```

The IPv6 address **2001:db8:100:1::10** should appear under the Ethernet adapter.

Result: A valid IPv6 address from the selected subnet was manually assigned and verified.